

IILM University, Gurugram	School:	School of Computer Science & Engineering	Year:	I	Course Name	<i>Foundations of AI & Automation</i>
	Program:	B.Tech. (CSE)	Semester:	I	Course Code	
					L-T-P-C	2-0-0-2
					Prerequisite Subject	None
Course Objectives		1. To introduce the fundamental concepts, history, and applications of Artificial Intelligence and automation.				
		2. To develop an understanding of basic AI techniques, logic-based reasoning, and automation systems with real-life applications.				
		3. To expose students to foundational tools and real-world applications of AI and automation, encouraging curiosity and innovation in intelligent system design				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To understand and describe the scope, domains, and evolution of AI and automation technologies.				
CO2		To apply basic logic and reasoning techniques for solving AI-related problems.				
CO3		To understand components, structure, and functioning of basic robotic and automated systems.				
CO4		To analyze current applications and emerging trends in AI and robotics, along with their ethical and societal implications.				
Course Description		This course provides an introductory overview of Artificial Intelligence and automation, covering fundamental concepts, techniques, and real-world applications. Students will learn about intelligent agents, basic problem-solving methods, and emerging AI trends. The course aims to build a strong foundation for further studies in intelligent systems and automation technologies.				
Detailed Syllabus						
Unit	Topics				CO No.	Number of lectures
Unit 1: Introduction Artificial Intelligence	Definition and scope of Artificial Intelligence (AI), Historical context and evolution of AI-Turing Test and Chinese room argument, Problem reduction State space search: Problem-solving algorithm, Major Components of AI and Search Process				CO1	6

Unit 2: Intelligent Agents and Problem Solving	Introduction to Agents and environment, nature of environment and agents, problem solving agent, Search Algorithm: Brute Force Search, Depth First Search, Breadth First Search, Heuristic Search Algorithm: Hill Climbing Algorithm, Best First Search, A and AO* Algorithm, Knowledge Representation	CO2	7
Unit 3: Introduction to Robotics and Automation	Introduction to robotics, Basic Concepts, History and Classification, Types of Robots and Automation Systems like Industrial robots, mobile robots, humanoid robots, Smart appliances, Key Components: Sensors, Actuators, Controllers, Introduction to Control Systems, Automation in Manufacturing, Smart Homes, and Agriculture	CO2, CO3	9
Unit 4: Applications of Robotics and Future Trends	Aerial Robotics: Introduction to Drones (UAVs), Components: Frame, motors, propellers, flight controller, Flight principles: Lift, thrust, drag, Applications: Surveillance, mapping, delivery. Emerging Trends in Robotics: Swarm Robotics, Collaborative Robots (Cobots), Soft Robotics, Human-Robot Interaction (HRI)	CO3, CO4	9

Text Books :

1. [Machine Learning, Tom Mitchell, McGraw Hill, 1997.](#)
2. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall, Third Edition, 2009.
3. "Introduction to Robotics: Mechanics and Control" by John J. Craig

Reference Books :

1. Bratko, I. (2011). *Prolog: Programming for Artificial Intelligence* (4th ed.). Addison-Wesley Educational Publishers Inc.
2. Janakiraman, V. T., Sarukesi, K., & Gopalakrishnan, P. (2001). *Foundations of Artificial Intelligence and Expert Systems*. New Delhi: Macmillan India Ltd.
3. Introduction to AI Robotics" by Robin R. Murphy
4. Drone Fundamentals: Theory and Practice for Drone Pilots by David G. Price

Online Resources:

1. <https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python>
2. <https://ocw.mit.edu/>

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination
Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

[illegible]